

LARSEN ICE SHELF, Antarctica

WMO: 892620

Lat: 67.013S

Lon: 61.470W

Elev: 45

StdP: 100.79

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-41.9	-39.5	N/A	N/A	N/A	N/A	N/A	N/A	16.7	-18.6	14.7	-17.8	1.5	250	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.5	5.2	N/A	3.5	N/A	2.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	10.6	9.1	DB	-44.5	9.4	2.5	2.0	-46.3	10.9	-47.7	12.0	-49.1	13.1	-50.9	14.6	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-14.9	-1.4	-4.9	-12.2	-20.5	-21.6	-25.3	-25.5	-23.6	-19.4	-13.7	-7.3	-2.6		
	DBStd	10.62	2.13	3.88	5.68	6.72	7.80	6.41	7.80	8.29	7.76	6.16	4.43	3.06		
	HDD10.0	9078	352	418	688	914	981	1059	1101	1042	881	733	518	389		
	HDD18.3	12120	610	652	947	1164	1239	1309	1359	1301	1131	992	768	647		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	3.7	3.4	3.6	3.6	3.4	3.1	3.5	3.9	4.0	3.9	4.1	3.8	3.7		
(15) Precipitation	PrecAvg	689	54	57	65	57	54	47	51	54	64	74	66	47		
	PrecMax	1016	114	109	96	122	129	88	109	95	101	134	120	109		
	PrecMin	434	12	33	36	17	30	20	16	15	35	45	24	14		
	PrecStd	184	26	19	16	24	21	19	20	19	18	23	26	22		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	5.1	3.1	1.9	2.1	0.3	-0.3	-1.2	1.7	3.4	5.1	7.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	5.5	2.9	1.0	-2.7	-0.9	-7.5	-3.0	-4.0	-0.7	1.4	3.0	4.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	3.2	1.6	-1.4	-7.5	-4.6	-12.4	-7.5	-6.6	-3.5	-0.8	1.5	3.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	2.1	0.3	-3.3	-11.3	-9.5	-16.3	-12.8	-10.0	-7.2	-3.2	-0.5	1.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.0	8.9	8.7	9.5	9.4	9.9	11.0	10.7	10.9	9.6	7.5		
		MCDBR	9.1	8.7	11.2	13.9	13.3	15.2	16.2	14.7	12.7	12.9	12.0	10.4		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.270	0.261	0.239	0.194	0.133	N/A	0.131	0.189	0.227	0.245	0.256	0.267		
	taud		2.279	2.291	2.298	2.181	1.914	N/A	1.901	2.120	2.197	2.187	2.209	2.233		
	Ebn at Noon		986	936	855	676	283	0	254	660	849	944	996	1003		
	Edh at Noon		114	97	73	42	11	0	10	43	78	107	121	124		
(44) All-Sky Solar Radiation	RadAvg		7.74	5.38	2.88	0.97	0.14	0.00	0.05	0.57	2.15	4.64	7.26	8.66		
	RadStd		0.23	0.15	0.08	0.04	0.01	0.00	0.00	0.03	0.07	0.10	0.21	0.26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page